

Separable Differential Equations



Solving by separation of variables

- 1 Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
 - 2 Solve $\frac{dy}{dx} = 2xy$, given $y(0) = 5$.
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Growth, decay, and applications

- 3 A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.
 - a Solve for $P(t)$.
 - b Find $P(20)$, to the nearest whole number.
- 4 Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a 20°C room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.